

courtship (see Figure 51.4). Pheromones are not limited to short-distance signaling, however. Male silkworm moths have receptors that can detect the pheromone from a female moth from several kilometers away (see Figure 50.6).

In a honeybee colony, pheromones produced by the queen and her daughters, the workers, maintain the hive's complex social order. One pheromone (once called the queen substance) has a particularly wide range of effects. It attracts workers to the queen, inhibits development of ovaries in workers, and attracts males (drones) to the queen during her mating flights out of the hive.

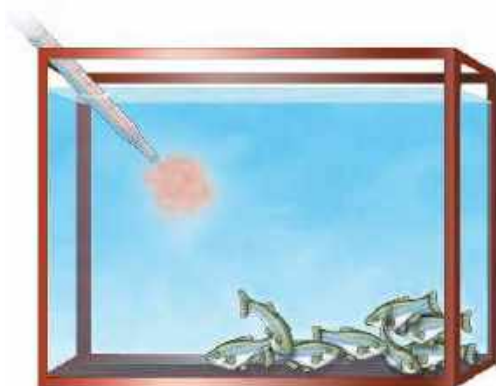
Pheromones can also serve as alarm signals. For example, when a minnow is injured, a substance released from the fish's skin disperses in the water, inducing a fright response in other minnows in the area. These nearby fish become more vigilant and often form tightly packed schools near the river or lake bottom, where they are safer from attack (Figure 51.6). Pheromones can be very effective at remarkably low concentrations. For instance, just 1 cm² of skin from a fathead minnow contains sufficient alarm substance to induce a reaction in 58,000 L of water.

So far in this chapter, we have explored the types of stimuli that elicit behaviors—the first part of Tinbergen's first question. The second part of that question—the physiological mechanisms that mediate responses—involves the nervous, muscular, and skeletal systems explored elsewhere in this unit: Stimuli activate sensory systems, are processed in the central nervous system, and result in motor outputs that constitute

▼ **Figure 51.6 Fathead minnows (*Pimephales promelas*) responding to the presence of an alarm substance.**



1 Minnows are widely dispersed in an aquarium before an alarm substance is introduced.



2 Within seconds of the alarm substance being introduced, minnows aggregate near the bottom of the aquarium and reduce their movement.

behavior. Thus, we are ready to focus on Tinbergen's second question—how experience influences behavior.

CONCEPT CHECK 51.1

1. If an egg rolls out of a greylag goose's nest, the mother goose will retrieve it by nudging it with her beak and head. If researchers remove the egg or substitute a ball during this process, the goose continues to bob her beak and head while she moves back to the nest. Explain how and why this behavior occurs.
2. **WHAT IF?** Suppose you exposed various fish species from the minnows' environment to the alarm substance from minnows. Thinking about natural selection, suggest why some species might respond like minnows, some might increase their activity, and some might show no change.
3. **MAKE CONNECTIONS** How is the lunar-linked rhythm of fiddler crab courtship similar in mechanism and function to the seasonal timing of plant flowering? (See Concept 39.3.)

For suggested answers, see Appendix A.

CONCEPT 51.2

Learning establishes specific links between experience and behavior

Some behaviors—such as a fixed action pattern, a courtship stimulus-response chain, or pheromone signaling—are performed by all individuals the same way each time. Behavior that is developmentally fixed in this way is known as **innate behavior**. Consider, for example, the construction of a spider's web. Although a spider's web is quite complex, web building is innate, not learned. Many other behaviors, however, vary considerably with experience.

► **A spider in family Theridiidae spinning a web**



Experience and Behavior

Tinbergen's second question asks how an animal's experiences during growth and development influence the response to stimuli. One informative approach to this question is a *cross-fostering study*, in which the young of one species are placed in the care of adults from another species in the same or a similar environment. The extent to which the offspring's behavior changes in such a situation provides a measure of how the social and physical environment influences behavior.

Certain mouse species have behaviors well suited for cross-fostering studies. Male California mice (*Peromyscus californicus*)